MILITARY SPECIFICATION

COUPLER, ANTENNA CU-351(*)/AR
and
COUPLER, ANTENNA CU-509(*)/AR

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

*1.1 Scope - The equipment covered by this specification shall provide automatic tuning of typical fixed aircraft antennas and efficient transfer of RF power from the 52 ohm output of a 50-150 watt transmitter into such antennas.

*1.2 Classification - The equipment covered by this specification shall be of two types and shall consist of the following items:

Items	Type Designation	Appl. Para.
Coupler, Antenna	CU-351(*)/AR	3.4.1
Coupler, Antenna	CU-509(*)/AR	3.4.2
Mounting	MT-1223(*)/AR	3.4.1.2
Mounting	MT-1660(*)/AR	3.4.2.2

1.3 Associated Equipment - The equipment shall operate with the following associated equipment, which shall not be supplied as part of this equipment.

(1) Transmitter	Any transmitter with a 52 ohm output impedance yielding between 50 and 150 watts output.
(2) Antenna	Fixed, aircraft.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

FSC 5985

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SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations

PUBLICATIONS

Bureau of Naval Weapons

SAR-300	Material Changes and Material Bulletins, Preparation of

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*2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and, the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall be first approved by the procuring activity.

*3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

*3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

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3.1.3 Finishes - Brass hardware, shafts, rod or tubing, and gears, may be used without a protective finish where the part is subject to wear.

3.1.4 Inductors -

3.1.4.1 Stability of Q - Every precaution shall be taken to maintain stability of Q in radio frequency inductors to insure proper performance of the equipment. The stability of Q requirements MIL-E-5400 need not apply provided the equipment satisfies all of the requirements for operation under Service Conditions as specified herein.

*3.1.5 Switches - In addition to the requirements of Specification MIL-E-5400, rotary switches mechanically positioned by other than manual means shall be mounted by attachment of the through bolts to standoffs and braced to meet the vibration and shock requirements of this specification. Bracing shall be effected by attachment of braces to the through bolts.

*3.2 General Requirements - This equipment shall fulfill all applicable design, workmanship and construction requirements of Specification MIL-E-5400 unless otherwise specified herein.

*3.2.1 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment: -

Temperature	Room ambient (30° ± 10° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in Specification MIL-E-5400 for class 1 equipment, unless otherwise specified herein.

*3.2.2.1 Altitude - The antenna coupler CU-351/AR shall operate without corona or flashover at altitudes up to 30,000 feet when used with the RT-311/ARC-38 and the long antenna as specified in 3.3.4. The CU-351/AR shall operate without flashover at altitudes up to 20,000 feet with either long or short antenna specified in 3.3.4 when used with a transmitter with a 50 ohm output impedence yielding between 50 and 150 watts output modulated 95%.

*3.2.3 Total Weight - The total weight of the CU-351/AR excluding cables shall not exceed 23 pounds. The total weight of the CU-509/AR excluding cables shall not exceed 35 pounds.

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3.2.4 Operational Stability - The equipment shall operate with optimum performance for 100 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

3.2.5 Operating Life -

3.2.5.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6 Interconnecting Cables - This equipment shall be capable of satisfactory operation using external wiring in accordance with Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished.

*3.2.7 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. Cables and that portion of the connectors which are to be attached to the cables shall not be supplied as part of the equipment.

*3.2.8 Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified herein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B"

Power above - 20° C	20 VA
Power below - 20° C	65 VA

(2) DC Power, 28 Volt, Category "B",

Current	3.5 Amps

(3) DC Power from Associated Transmitter

Operating Voltage Limits	225 to 265 volts
or	
Operating Voltage Limits	360 to 420 volts
Current	35 ma

3.2.8.1 Power Supplied by Associated Transmitter - The equipment shall operate properly when 250 volts or 400 volts is supplied by the associated transmitter. Provision shall be made for shorting the series dropping resistors within the equipment when 250 volts is supplied.

*3.2.9 Input Power Connection - Input power connections shall be in accordance with MS25245.

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3.2.10 Overload Protection - Overload protection for the equipment is provided in the associated transmitter equipment. No additional overload protection shall be required in this equipment.

3.2.10.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified, and shall resume normal tuning operations when the voltages return within limits and the associated transmitter is re-channelled.

*3.2.11 Standby Operating Provisions - The equipment shall operate in standby condition with any or all rated power except RF power applied continuously.

3.2.12 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

*3.2.13 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

*3.2.14 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

*3.2.15 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3 Performance

3.3.1 General - The performance requirements of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and service conditions.

*3.3.2 Frequency Range - The equipment shall be capable of proper operation with an associated transmitter over the frequency range of 2 to 23 Mc/s.

3.3.3 Types of Emission - The equipment shall operate properly under all types of emission.

*3.3.4 R.F. Terminations - When the coupler is suitably connected between the 52-ohm output of the associated transmitter and an aircraft antenna having an effective length between that of the long antenna and that of the short antenna listed below, it shall be possible to adjust the output circuits of the transmitter in such a manner that the final amplifier stage of the transmitter will operate at normal input power. The equipment shall operate properly when connected to a typical ungrounded, fixed wing aircraft antenna between 45 and 100 feet in length. For laboratory and bench test evaluation purposes, the equipment shall operate properly using dummy antennae with the following characteristics:

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Freq. mcs	Long Antenna R	Long Antenna X	Long Antenna Eff. %	Short Antenna R	Short Antenna X	Short Antenna Eff. %
2	3	-j200	30%	3	-j450	20%
3	4	-j10	75%	3.5	-j250	30%
4	7	+j100	50%	4	-j100	40%
5	20	+j350	50%	5	-j10	75%
6	200	+j1000	75%	6.5	+j100	50%
7	90	-j800	50%	10	+j250	50%
8	25	-j250	50%	20	+j400	60%
9	16	-j70	75%	50	+j700	75%
10	16	+j60	75%	500	+j2000	60%
11	25	+j250	75%	300	-j1000	60%
12	90	+j800	60%	50	-j500	60%
13	1000	-j2000	50%	25	-j300	60%
14	90	-j400	60%	15	-j200	60%
15	35	-j150	60%	10	-j80	60%
16	25	+j0	50%	10	+j60	75%
17	35	+j150	75%	15	+j200	75%
18	90	+j400	75%	25	+j300	75%
19	500	+j2000	50%	50	+j500	60%
20	200	-j600	50%	300	+j1000	50%
21	60	-j250	50%	2000	-j2000	50%
22	42	-j20	60%	300	-j700	50%
23	45	+j80	60%	100	-j400	50%

3.3.5 Maximum Duty Cycle - The equipment shall operate properly during all modes of operation except FSK when a maximum duty cycle of RF power ON for 5 minutes and OFF for 5 minutes is not exceeded. During the FSK mode of operation, the equipment shall operate properly with a maximum duty cycle of 100%.

NOTE: Paragraph 3.3.5 includes requirements for FSK operation which were not included in previous editions of this specification. Contractors are advised that equipments produced in accordance with prior specification editions may not include the specified FSK capability but that such exclusion from said equipment shall not, in any way, be construed as a waiver to the requirements specified herein.

*3.3.6 Efficiency - When the equipment is utilized in the manner described in 3.3.4 the minimum efficiency of RF power transfer from the associated transmitter to the antenna, for the purpose of laboratory test, shall be derived by dividing the RF power delivered to the dummy antenna by the RF power input to the antenna coupler. The efficiencies listed in 3.3.4 are the minimum allowable efficiencies for the given antenna characteristics.

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*3.3.7 Tune-up Time - When the equipment is utilized in the manner described in 3.3.4 and the associated transmitter generates a radio frequency carrier wave on any frequency within the 2 to 23 mc range, all circuits of this equipment requiring tuning shall be tuned to resonance and coupled to optimum by automatic means. The VSWR presented to the 52 ohm output of the associated transmitter shall not exceed 1.7 to 1. The time elapsing between the time the transmitter commences to generate a radio frequency carrier wave on the selected frequency channel and the time that the output circuits are tuned and coupled shall not exceed 25 seconds under the standard conditions nor 30 seconds under the service conditions specified herein.

*3.3.8 Automatic Keying and Holding - The equipment shall provide automatic keying and holding of the associated transmitter until the antenna has been tuned, at which time this equipment shall release the transmitter keying circuit. The equipment shall provide for selection of this function between one of two choices as listed below.

(1) Automatic Keying - The equipment shall key the transmitter any time the transmitter is channelled to a new frequency and the key shall be automatically held until the antenna has been tuned.

(2) Radio Silence Keying - The equipment shall key the transmitter any time the transmitter has been channelled to a new frequency followed by operation of the transmitter key by the operator and the key shall be automatically held until the antenna has been tuned.

3.3.9 Indication - The equipment shall provide circuits that may be used externally to indicate visually or aurally that the equipment is in the process of automatically tuning or has completed the tuning process.

*3.3.10 Preliminary Tests - The equipment, including the associated transmitter, may be operated and channelled up to and including three times during the specified warm-up period to determine satisfactory operation of the equipment before starting detailed tests.

*3.3.11 Service Performance - The output of the antenna coupler measured under service conditions shall be within plus or minus 2 db of the power output measured during the reference run conducted on the equipment.

3.4 Detail Requirements -

*3.4.1 Coupler, Antenna CU-351(*)/AR - The antenna coupler shall meet the following requirements:

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*3.4.1.1 Function - The coupler, when properly interconnected with an associated transmitter and energized from the power sources specified under 3.2.8, shall automatically and efficiently transform the impedance of aircraft antennas to 52-ohm resistances with a VSWR not to exceed 1.7 to 1.

*3.4.1.2 Form Factor and Mounting - The case and mounting shall be in accordance with Specification MIL-C-172 except that the case dimensions shall not exceed 13-7/8 by 10-3/8 by 7-13/16 inches and the mounting base dimensions shall not exceed 11-1/4 by 10-1/4 by 2 inches. The Mounting MT-1223(*)/AR shall be suitable for attachment to a plane surface in an aircraft and for supporting the coupler which shall be removable without the use of tools.

3.4.1.3 Contents - The coupler shall contain antenna matching elements, antenna transfer relay, vacuum tubes, electro-mechanical tuning devices, associated relays and control circuits, automatic keying circuits and shall be contained in a single non-pressurized case.

*3.4.1.4 Weight with Mounting - The weight, including Mounting MT-1223(*)/AR, shall not exceed 23 pounds.

*3.4.1.5 External Electrical Connection - The coupler shall provide the following connection facilities:

(1) A multiple contact receptacle for connection to the associated transmitter circuits, receiver disabling circuits, and to the primary power sources.

(2) A suitable antenna connector for connection to the aircraft antenna.

(3) A suitable coaxial connector for connection to the output circuits of the associated transmitter.

(4) A suitable coaxial connector for connection of auxiliary receivers to the aircraft antenna through a transfer relay included in this coupler. Means shall therefore be provided for transfer of the aircraft antenna from the output of the antenna coupler to the input of auxiliary receivers whenever the associated transmitter is not keyed. A separate means shall be provided internally whereby the input of the auxiliary receivers and the antenna may be grounded. The control voltage for the latter means shall be from an external source of 27.5 volts and shall be supplied to the coupler through contacts of the receptacle described under 3.4.1.5 (1).

3.4.1.6 Internal Electrical Connections - Means shall be provided to effect the following changes in internal connections by means of a conventional, straight-tipped, screwdriver only:

(1) Change in power supplied by the associated transmitter per 3.2.8.1.

(2) Change in keying requirements per 3.3.8..

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3.4.1.7 Controls - All semi-fixed adjustable devices provided for tracking, alignment, gain, etc., shall be readily accessible when the unit is removed from its case. If tools other than a conventional straight-lipped screwdriver are required for manipulation of these devices, such tools shall be furnished and clipped to the unit inside the case. The unit shall not require external manual controls for operation other than those provided with the associated transmitter.

3.4.2 Coupler, Antenna CU-509(*)/AR - The antenna coupler shall meet the following requirements:

*3.4.2.1 Function - The coupler, when properly interconnected with an associated transmitter and energized from the power sources specified under 3.2.8, shall automatically and efficiently transform the impedance of aircraft antennas to 52-ohm resistances with a VSWR not to exceed 1.7 to 1.

3.4.2.2 Form Factor and Mounting - The case and mounting shall be in accordance with Specification MIL-C-172 except that the case dimensions shall not exceed 13-7/8 by 12-1/8 by 8-5/8 inches and the mounting base dimensions shall not exceed 15-1/4 by 13 by 2-1/4 inches. The Mounting MT-1660(*)/AR shall be suitable for attachment to a plane surface in an aircraft and for supporting the coupler which shall be removeable without the use of tools.

3.4.2.3 Contents - The coupler shall contain antenna matching elements, antenna transfer relay, vacuum tubes, electro-mechanical tuning devices, associated relays and control circuits, automatic keying circuits, and shall be contained in a single pressurized case.

*3.4.2.4 Weight with Mounting - The weight of the coupler, including Mounting MT-1660(*)/AR, shall not exceed 35 pounds.

3.4.2.5 External Electrical Connection - The coupler shall provide the following connection facilities:

(1) A multiple contact receptacle for connection to the associated transmitter circuits, receiver disabling circuits, and to the primary power sources.

(2) A suitable antenna connector for connection to the aircraft antenna.

(3) A suitable coaxial connector for connection to the output circuits of the associated transmitter.

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(4) A suitable coaxial connector for connection of auxiliary receivers to the aircraft antenna through a transfer relay included in this unit. Means shall, therefore, be provided for transfer of the aircraft antenna from the output of the antenna coupler to the input of auxiliary receivers whenever the associated transmitter is not keyed. A separate means shall also be provided internally whereby the input of the auxiliary circuits may be independently disconnected from the antenna input and grounded. The control voltage for the latter means shall be from an external source of 27.5 volts and shall be supplied through the contacts of the receptacle described under 3.4.1.5(1).

3.4.2.6 Internal Electrical Connection - Means shall be provided to effect the following changes in internal connections by means of a conventional straight-tipped, screwdriver only:

(1) Change in power supplied by the associated transmitter per 3.2.8.1.

(2) Change in keying requirements per 3.3.8.

3.4.2.7 Controls - All semi-fixed adjustable devices provided for tracking, alignment, gain, etc., shall be readily accessible when the unit is removed from its case. If tools other than a conventional straight-tipped screwdriver are required for manipulation of those devices, such tools shall be furnished and clipped to the unit inside the case. The unit shall not require external manual controls for operation other than those provided with the associated transmitter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.7. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to an, and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

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(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its

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various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

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4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

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(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Reconditioning of Acceptance Test Samples - Each equipment which has been subjected to acceptance tests (Sampling, special and life tests) shall be reconditioned as follows:

(1) On completion of the Sampling, Special or Life Test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.7 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.10 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specification - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.12. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Definitions

6.5.1 Tuning Time - The tuning time is the period of time elapsing between the time the transmitter commences to deliver r-f power to the antenna coupler on the selected frequency channel and the time the antenna coupler delivers maximum power to the antenna.

6.5.2 Life Cycles - To determine life in terms of cycles of operation, the cycling frequency of the equipment is considered as once every five minutes.

6.6 An asterisk preceding a paragraph number denotes a change in that paragraph from the original specification.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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